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Markets Prep

Bill Hrusa (CMU Math)

Cell 412-965-1159

wh15@andrew.cmu.edu

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• What is Finance?

→ management of money

→ allocation and decision making of how to invest money

→ Study of risk and return

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My favorite definition is

" Allocation of resources over time and various assets. "

• What is money?

→ Store of value

→ Standardized medium of exchange

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• Why do people participate (trade) in financial markets?

- hedge (protect), insurance
- Save (defer purchasing power)
- Speculate (try to make a profit)
- facilitate economic activity, provide liquidity (market maker)

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• What is a security?

(Practical and Legal Defns)

→ Any tradable financial instrument.

→ A document (possibly electronic) which gives holder a right to a ~~share~~ financial claim.

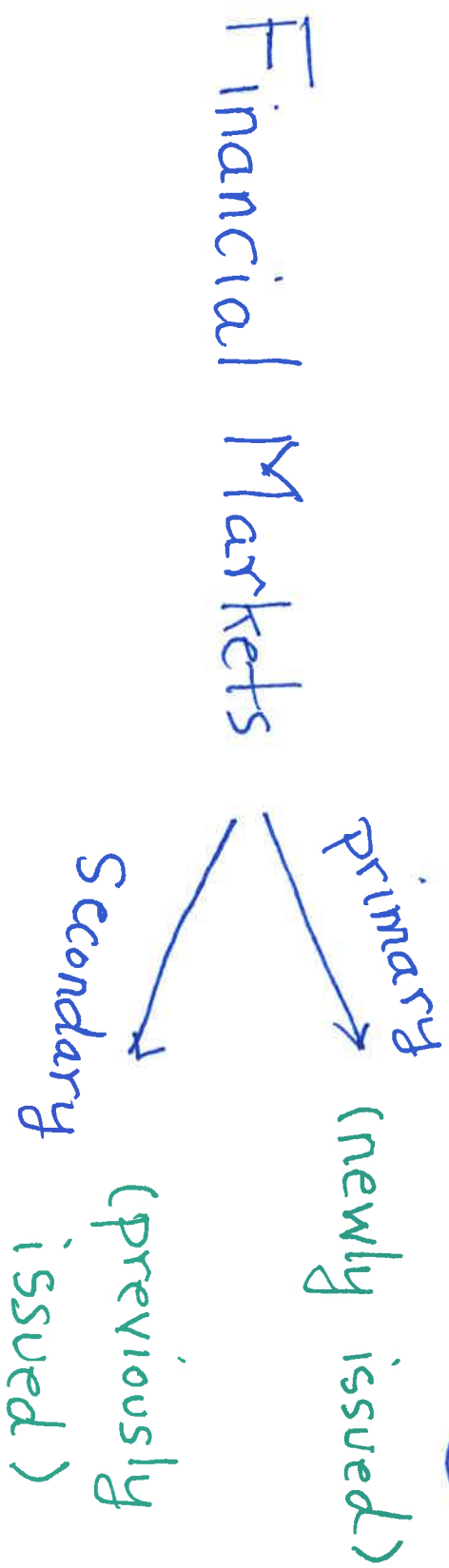
Classes of Securities ⑥

Equities (e.g. common stocks)

Debt Securities (Bonds)

Derivative Securities (payoff based on value of some "financial" variable).

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Exchange Traded

Over The Counter (OTC)

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Asset Classes

1. Equities
2. Rates Products (Fixed Income)
3. Currencies
4. Commodities

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Sectors of Stock Market

Utilities, Technology, Financials,
Industrials, Consumer Staples,
Consumer Discretionary, Energy,
Materials, Real Estate,
Health Care, Telecom

Market Capitalization

Mega Caps: 200B and up

Large Caps: ~10B - 200B

Mid Caps: ~2B - 10B

Small Caps: ~300M - 2B

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Banks

Commercial Banks (Retail Banks): Take Deposits and Lend Money

Investment Banks: Connect People who have ideas but lack capital with investors who have excess capital.

Central Banks: Governments

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Regulations

Financial Crises generally lead to new regulations

→ Great Depression (1929, 1930's)

→ 2008 Financial Crises

Glass - Steagall Act 1933 (Separated Commercial & Investment Banking Activities) Repealed in 1999

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Dodd-Frank (Following 2008 Financial Crises)

Volcker Rule (Prevents Investment Banks from Proprietary trading)

US Treasuries are Exempt!

Some Key Stocks Ticker Symbol

Current Share Prices in \$

Apple	AAPL	146	[2.42T]
Microsoft	MSFT	289	[2.18T]
Amazon	AMZN	3,345	[1.69T]
Alphabet	GOOG	2,741	GOOGL 2,715 [1.82T]
Face Book	FB	364	[1.025T]
Netflix	NFLX	521	
Zoom	ZM	383	
Boeing	BA	231	
Delta	DAL	40.32	

Tesla	TSLA	699
Exxon Mobile	XOM	57.86
Johnson & Johnson	JNJ	173
Pfizer	PFE	45.07
Moderna	MRNA	414
Berkshire - Hathaway	BRK-A	430,160
	BRK-B	286
Verizon	VZ	55.22
Ford	F	13.80

Investment Banks

Bank	Market Cap	Share Price
JPMorgan Chase (JPM)	471 B	158
Bank of Amer. (BAC)	338 B	40
Wells Fargo (WFC)	200 B	49
Morgan Stanley (MS)	182 B	100
Royal Bank of Canada (RY)	147 B	103
Citi (C)	144 B	71
Goldman Sachs (GS)	134 B	398
BNP-Paribas (BNPQF)	80 B	62
Barclays (BCS)	44 B	10
Deutsche Bank (DB)	26 B	13
Credit Suisse (CS)	26 B	11

Stock Indexes

DOW	iPhone by default) 30 Large US Companies	35,209
S&P 500	(AGSPC on iPhone)	4,437
NASDAQ	(AIXIC on iPhone)	14,836
Russell 2000	(ARUT on iPhone)	2,248

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DOW (Price Weighted)

Other 3 weighted by Market Cap

S&P 500 80% of US Market

NASDAQ Tech heavy, includes

foreign companies
National Assn. Sec. Dealers automated quotations

Russell 3000

~~Russell 3000~~ Russell 1000 (Lgst 1000 of 3000)

Russell 2000 (Sm1st 2000 of 3000)

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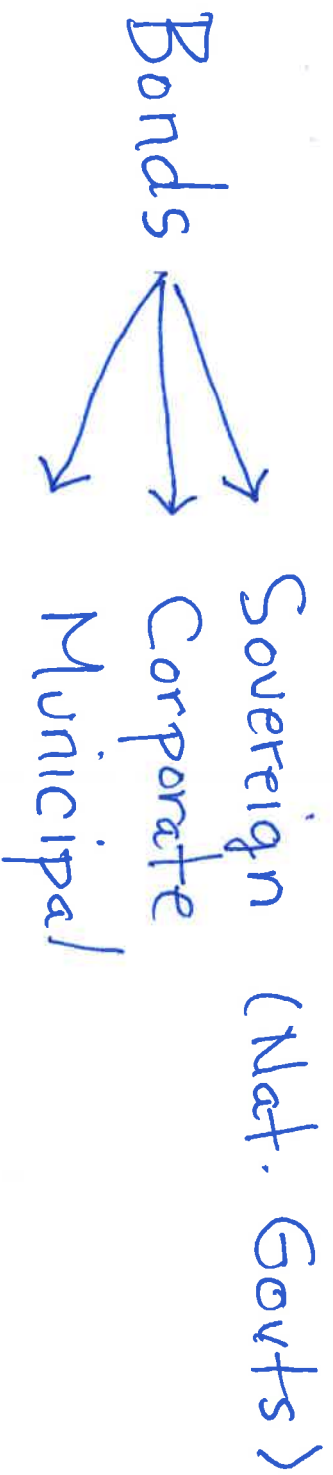
Indexes based on Share
Price (ignore dividends)

Typical Dividends on S&P
Stocks $\sim 2\%$

VIX (17.10% today)

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Rates Products



Focus on US Treasuries.
with Fixed Payment

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Zero Coupon Bonds

Face Value F , Maturity T

Single Payment of F at T

Coupon Bonds

Face Value F , Maturity T
annual coupon rate g .

Make Coup payments every 6
months of amount $F \frac{g}{2}$,

plus return F at T

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T-Bills : ZCBs maturities 1m, 3m
6m, 1yr

T-Notes Coup Bonds $2 \leq T \leq 10$
($T=2, 3, 5, 7, 10$ when issued)

T-Bonds Coup Bonds $T > 10$
($T=20, 30$)

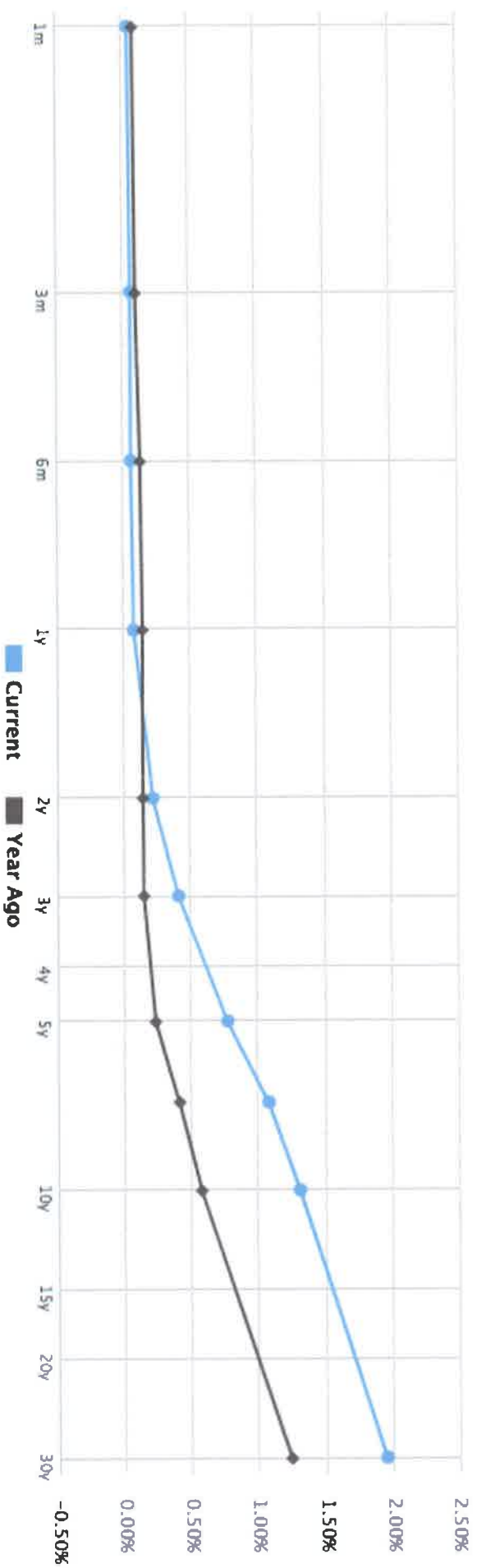
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Current Treasury Rates

3 m	.06%	(6 bp)
6 m	.06%	(6 bp)
1 yr	.09%	(9 bp)
2 yr	.21%	
5 yr	.77%	
10 yr	1.31%	
20 yr	1.85%	
30 yr	1.94%	

100 basis
points =
1 percentage
points
13 basis pts
= .13%
= .0013

Yield Curve



Swiss Franc

Currencies

USD

$$1 \text{ EUR} = 1.175 \text{ USD}$$

$$1 \text{ GBP} = 1.385 \text{ USD}$$

$$1 \text{ CHF} = 1.088 \text{ USD}$$

$$1 \text{ USD} = 110.24 \text{ JPY}$$

$$1 \text{ USD} = 6.485 \text{ CNY}$$

$$1 \text{ USD} = 1.257 \text{ CAD}$$

$$1 \text{ USD} = 1.359 \text{ AUD}$$

$$1 \text{ BTC} = 45,589$$

$$1.175 = \text{EUR/USD}$$

$$(1 \text{ USD} = .9191 \text{ CHF})$$

Look up

USDx

"Dixie"

Higher USDx

→ Stronger Dollar

Commodities

Energy: Crude Oil
Oil Products, Natural Gas,
Electricity
Precious Metals: Gold, Silver

Look UP
Lumber Prices

Industrial Metals
Copper, Aluminum, Steel

Agricultural: Sugar, Soy Beans, Cotton

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Crude Oil: WTI (West Texas Intermediate)
Futures 65.61 per barrel
(Barrel = 42 gals)

Brent Crude Futures 68.19
per barrel

Gold 1,732 per troy ounce
Silver 23.48 per troy ounce